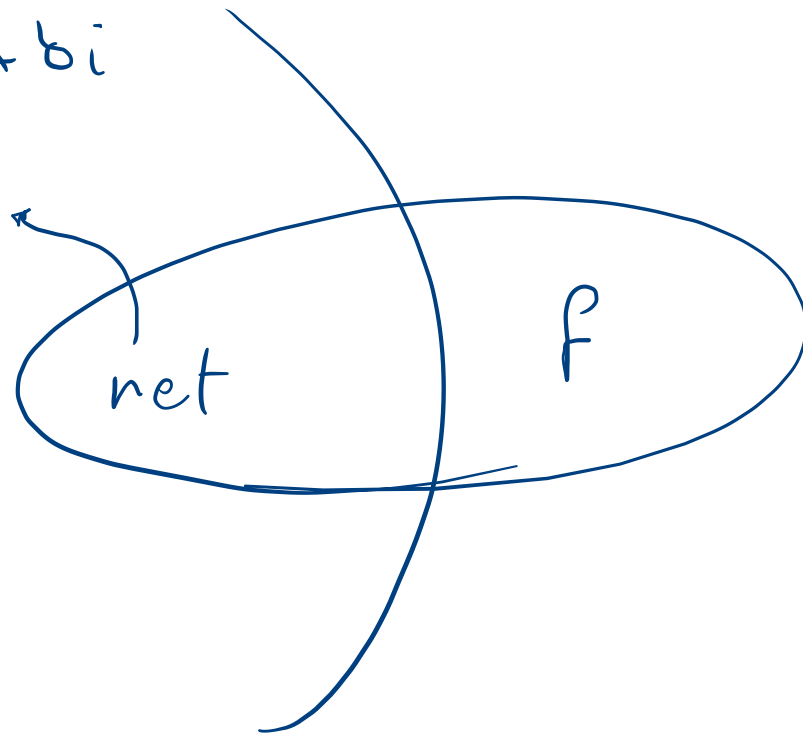


Deep learning

↳ neural nets.

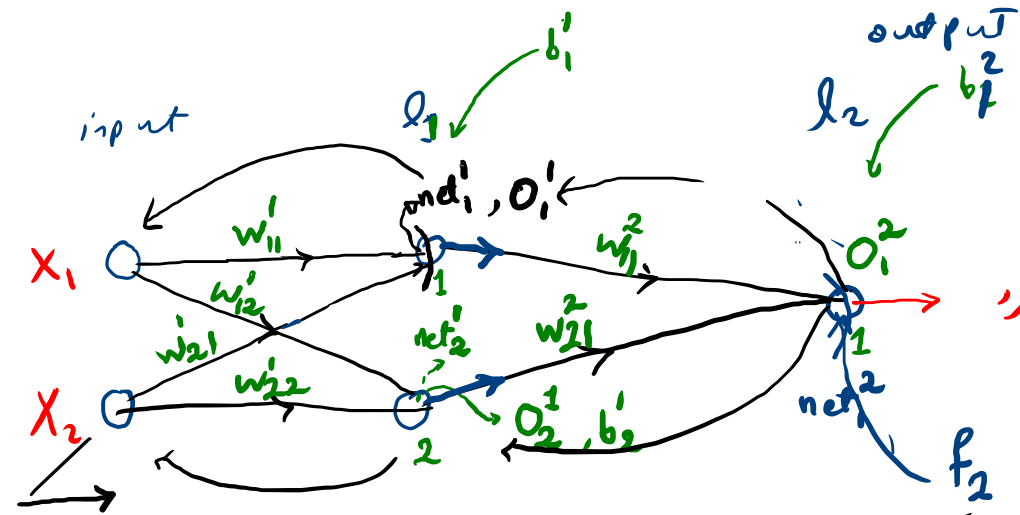
α

$$\text{net} = \sum w_i x_i + b_i$$



سن فرد : X_1

سن فرد : X_2



نتیجه به تیر فرد



Forward !
Feed Forward neural net

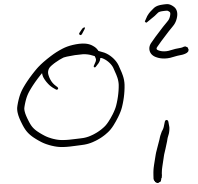
final output

$$net_1' = w_{11}'X_1 + w_{21}'X_2 + b_1' \rightarrow O_1' = f_1(net_1')$$

$$net_2' = w_{12}'X_1 + w_{22}'X_2 + b_2' \rightarrow O_2' = f_1(net_2')$$

$$net_1^2 = w_{11}^2 O_1' + w_{21}^2 O_2' + b_1^2 \rightarrow O_1^2 = f_2(net_1^2)$$

FFNN



$$O = f(\text{net})$$

↳ Activation Function —

range $[-1, 1]$

$[-1, +1]$

↗ binary classification

$$1) \quad \underline{\text{Sigmoid}} = \sigma = \frac{1}{1 + e^{-\text{net}}} = \sigma(x)$$

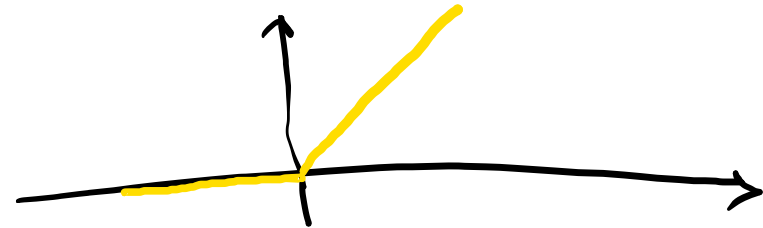
$f(\text{net})$
↳

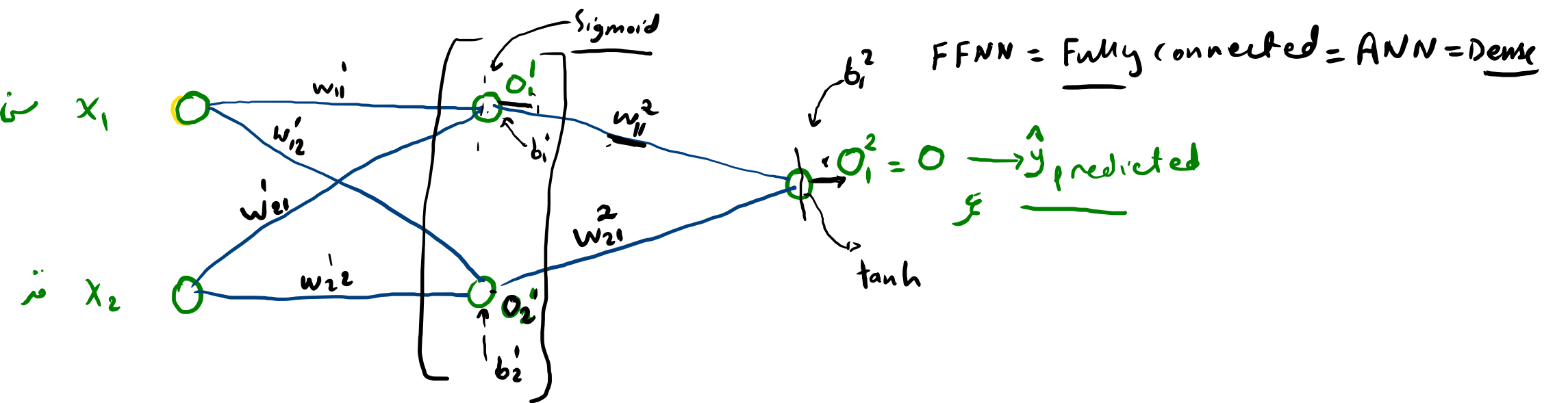
2,
$$\text{Softmax} = \frac{e^{-\text{net}}}{\sum e^{-\text{net}_i}} \rightarrow \text{multi-class classification.}$$

3,
$$\text{tanh} = \frac{e^{\text{net}} - e^{-\text{net}}}{e^{\text{net}} + e^{-\text{net}}}$$

4,
$$\text{Relu} = \begin{cases} 0 & \text{net} \leq 0 \\ \text{net} & \text{net} > 0 \end{cases}$$

$\text{net} \leq 0 \rightarrow \text{Relu}(\text{net}) = 0$
 $\text{net} > 0 \rightarrow \text{Relu}(\text{net}) = \text{net}$

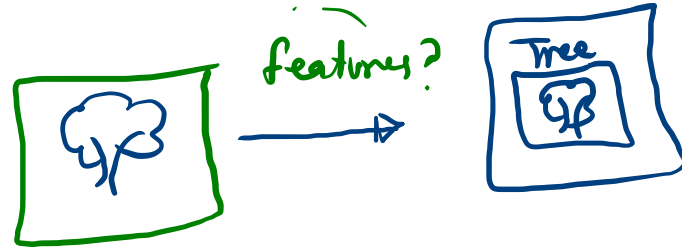
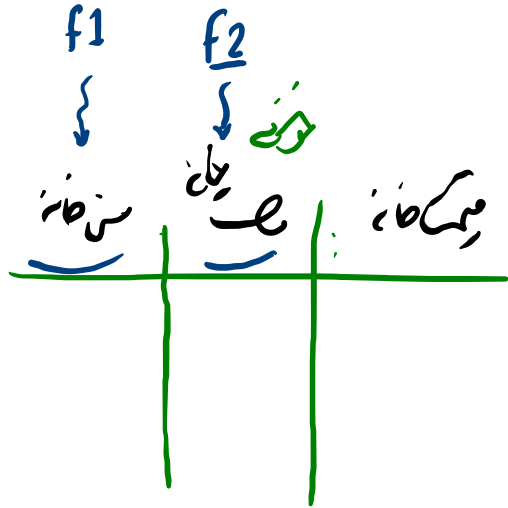




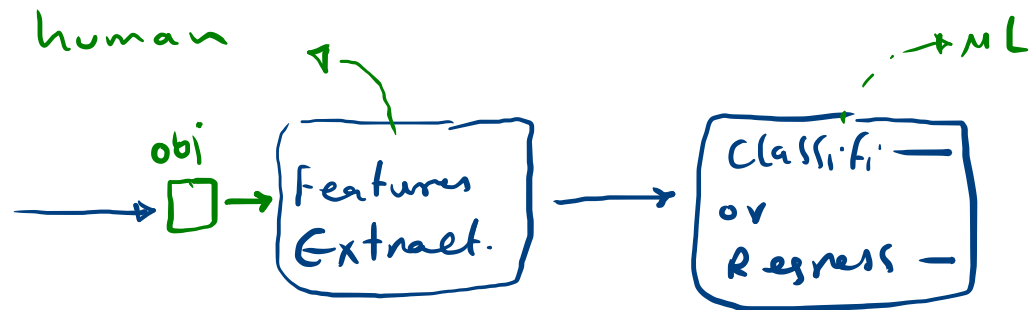
$$\begin{aligned}
 \text{net}_1^1 &= w_{11}^1 x_1 + w_{21}^1 x_2 + b_1 \rightarrow O_1' = \sigma(\text{net}_1^1) \\
 \text{net}_2^1 &= w_{12}^1 x_1 + w_{22}^1 x_2 + b_2 \rightarrow O_2' = \sigma(\text{net}_2^1) \quad \text{Layer 1} \\
 \text{net}_1^2 &= w_{11}^2 O_1' + w_{21}^2 O_2' + b_1^2 \Rightarrow O_1^2 = \tanh(\text{net}_1^2) \quad \text{Layer 2}
 \end{aligned}$$

$O_1^2 = ?$

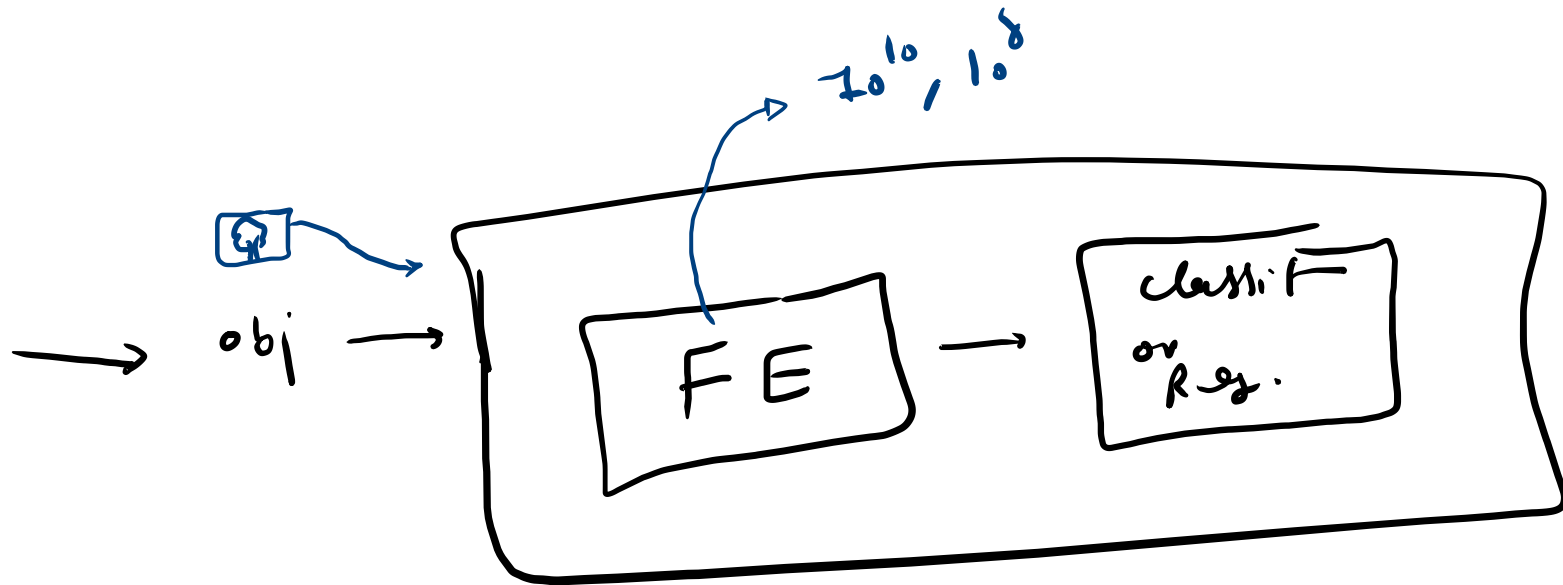
عاماً محدود سرعت سے کام لیتے ہیں، جبکہ ڈیپ لیئر نیورل نیٹ ورکس
 [classical ml] [Deep]

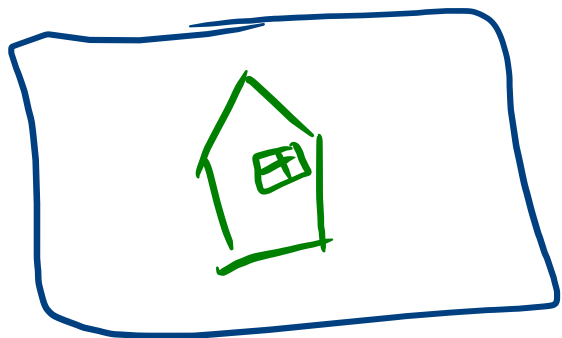


classical ml



Deep learning

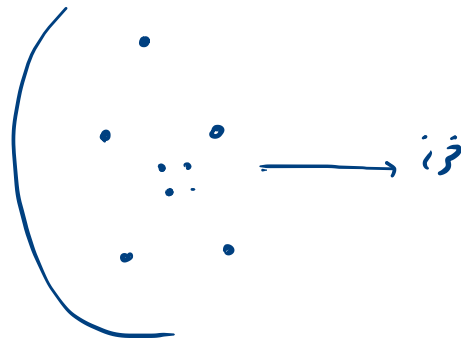




edges.



shape



۱. نوع ۱

FFNN
CNN
RNN
LSTM
GRU
...

۲. نوع ۲

۲. تعداد ۲۰۰

۴. حضرت اسماعیل علیہ السلام

طالع مریدت از پانزده چهره عظمیٰ — طالع مرید (انفـ بدل) —

x_1 x_2 y $\rightarrow y_{true} / y_{actual}$

x_1	x_2	y
$\square$	$\square$	y_1

$$\begin{array}{cc}
 x_1 & 0 \\
 x_2 & 0
 \end{array}
 \begin{array}{c}
 \text{---} \\
 \times \\
 \text{---}
 \end{array}
 \begin{array}{c}
 l_1 \\
 0 \\
 0
 \end{array}
 =
 \begin{array}{c}
 \text{نرخ} \\
 \text{میانگین مربعات} \\
 \text{NN} \\
 \text{میانگین مربعات}
 \end{array}
 \begin{array}{c}
 l_1 \\
 0 \rightarrow 0 : y_{predicted}
 \end{array}$$

$$mse = \frac{1}{2} \sum (y_{true} - y_{pred})^2$$

۱. محفہ کی سب سے زیادہ تیار اور مشہور - کلاسیک ، - Reg

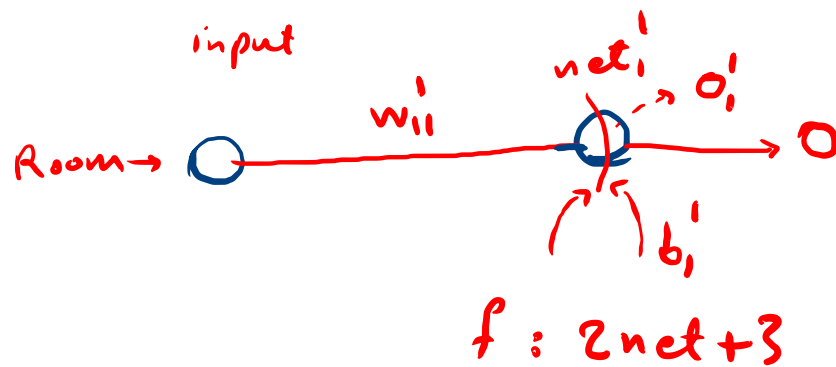
classical
ml

Deep Learning

Reg: mse - mae, RMSE
cls: cross entropy / prec
Recall
F1

DL

room	house-price
2	50



$$\begin{cases} w_{11} = 2 \\ b_1^1 = 2 \end{cases}$$

S1 : Room : 2

$$net_i^1 = w_{11} \overset{2}{x_1} + b_1^1 = 2w_{11} + b_1^1 = \underbrace{2 \times 2}_4 + 2 = 6$$

$$\rightarrow f = 2(6) + 3 = 15 \rightarrow y_{\text{predicted}}$$

$$mse = \frac{1}{2} \sum (y_{\text{pred}} - y_{\text{act}})^2 = \frac{1}{2} (15 - 50)^2 = \frac{1}{2} (35)^2 = \dots$$

$y_{\text{pred}} : 15$

$y_{\text{act}} : 50$

loss function.

۲. محاسبهٔ این آمار تابع زیان، به کمک تغییر در پارامترها، در جهت کاهش

تابع زیان
loss function
ی داریم!

روش / الگوریتم

Gradient Descent

گرادیان نزول

The End